Groundswell · Problem 2 · Components map

# A week of threads. One Friday brief.

How Groundswell turns live developer discussions into the brief Sumit takes to sprint planning. Code counts every number, Gemini Flash reads and explains, and Sumit decides what matters.

LIVE AT
devpulse-sable-seven.vercel.app

FROM 4 communities — TO 1 brief

| WHO / STOP | Trigger<br>What starts it | Input<br>What goes in | Context<br>What it knows | Processing<br>What happens | Output<br>What comes out |
|---|---|---|---|---|---|
| **Sumit**<br>PM, code review automation startup | SETS WHAT TO WATCH<br>Adds, edits or pauses topics and keywords on **/topics** (admin passcode). No code changes. Can press **Run now** for a live demo. | | ASKS ANYTIME<br>“Ask the Pulse”: a question is answered only from stored discussions, with links. Nothing relevant means “not enough data”. | HUMAN DECISION<br>**Approves new topics**<br>The radar suggests pain points no topic covers. Sumit taps **Approve & monitor** or **Reject**. Rejected ideas never come back. | HUMAN DECISION<br>**Takes it to the sprint**<br>Reads the brief before the afternoon engineering sync, opens the evidence, and chooses what goes into Monday’s sprint planning. |
| **Groundswell app**<br>Next.js on Vercel | STARTS THE RUN<br>The cron route checks the **CRON_SECRET**; Run now checks the admin cookie. Refuses to start if a run is already going. | COLLECTS & CLEANS<br>4 adapters in parallel, last 7 days, keyword match on title + body. Rejects off-site links, dedupes by id and URL, keeps the top 300. A failed source marks the run partial; the run continues. | COUNTS LAST WEEK FAIRLY<br>Smart baseline: Hacker News exact-phrase matches, this week vs the 7 days before. Under 5 in total → **“too few to compare”**, never a fake %. | SCORES IN CODE<br>Mentions, unresolved %, severity, trend. **Opportunity Score** = frequency × growth × unresolved × severity, top topic = 100. Every factor stored for “Why this score?”.<br><br>CHECKS THE AI<br>Turns evidence ids into real thread links, strips any URL Gemini typed, flags numbers not in its input. Radar ideas need ≥ 2 real threads. | SHOWS THE ANSWERS<br>Overview answers the three questions: **most discussed** (with counts), **up or down** vs last week, **still unresolved**. Plus Trends, Opportunities and a searchable Evidence feed. |
| **Gemini**<br>Flash, structured JSON | | | RECEIVES<br>Posts in batches of 25 plus the topic list, from the server only. Post text is marked as untrusted, so instructions inside it are ignored. | CLASSIFIES EACH POST<br>Topic, “other” or irrelevant · one-line pain point · unresolved + reason · sentiment · severity 1–5 · competitors named. Checked against a schema; retried once if invalid.<br><br>SPOTS NEW PAIN<br>Clusters “other” posts into up to 4 suggested topics with keywords and evidence. | EXPLAINS, NEVER COUNTS<br>Writes the brief’s commentary, citing evidence ids only. Tables and numbers come from code. Names likely sales objections with a suggested response. |
| **Supabase**<br>Postgres database | TOPICS<br>Monitored topics and their keywords. Editable from the app, so the watch list evolves with the product. | RUNS, POSTS<br>One row per run with per-source status. Each post keeps source, id, URL, published and retrieved time. | READ BY THE SERVER ONLY<br>Row-level security: the public key can only read. Writes need the service-role key on the server. Full-text index powers Ask. | TOPIC STATS<br>Counts, trend and every score component per topic per run, so this week can be compared with last week. | BRIEFS, SUGGESTIONS<br>The brief, opportunities, objections and email status; suggested topics waiting for Sumit. |
| **Outside services**<br>Live APIs, cron, email | VERCEL CRON<br>Every **Friday 08:00 IST** (02:30 UTC) it calls the weekly run. | 4 PUBLIC COMMUNITIES<br>Hacker News (Algolia) · GitHub Issues · Stack Overflow · DEV.to. Live data only, nothing mocked. | | | RESEND EMAIL<br>Delivers the brief to Sumit’s inbox the same morning, with links to every cited thread. |

### Code counts
Every number, trend and score is computed in code from stored data. Gemini cannot change one.

### Gemini reads and explains
It classifies posts and drafts the commentary, citing real threads. It never invents links.

### Sumit decides
He chooses what to monitor and what reaches the sprint. The tool stops at evidence.

KEY

- A person
- Code, hard rules (no AI)
- Database
- Human decision
- Gemini Flash (AI)
- Outside service

## The journey, from Friday 8:00 to Monday’s sprint

01 Cron fires Friday 8:00 IST
02 Load topics from Supabase
03 Collect from 4 communities
04 Gemini classifies each post
05 Code counts, trends, scores
06 Gemini drafts the brief
07 Code checks links & numbers
08 Emailed + on the dashboard

BEFORE THE SYNC

## Sprint-ready.

Most discussed, rising or falling, still unresolved, each with evidence links.